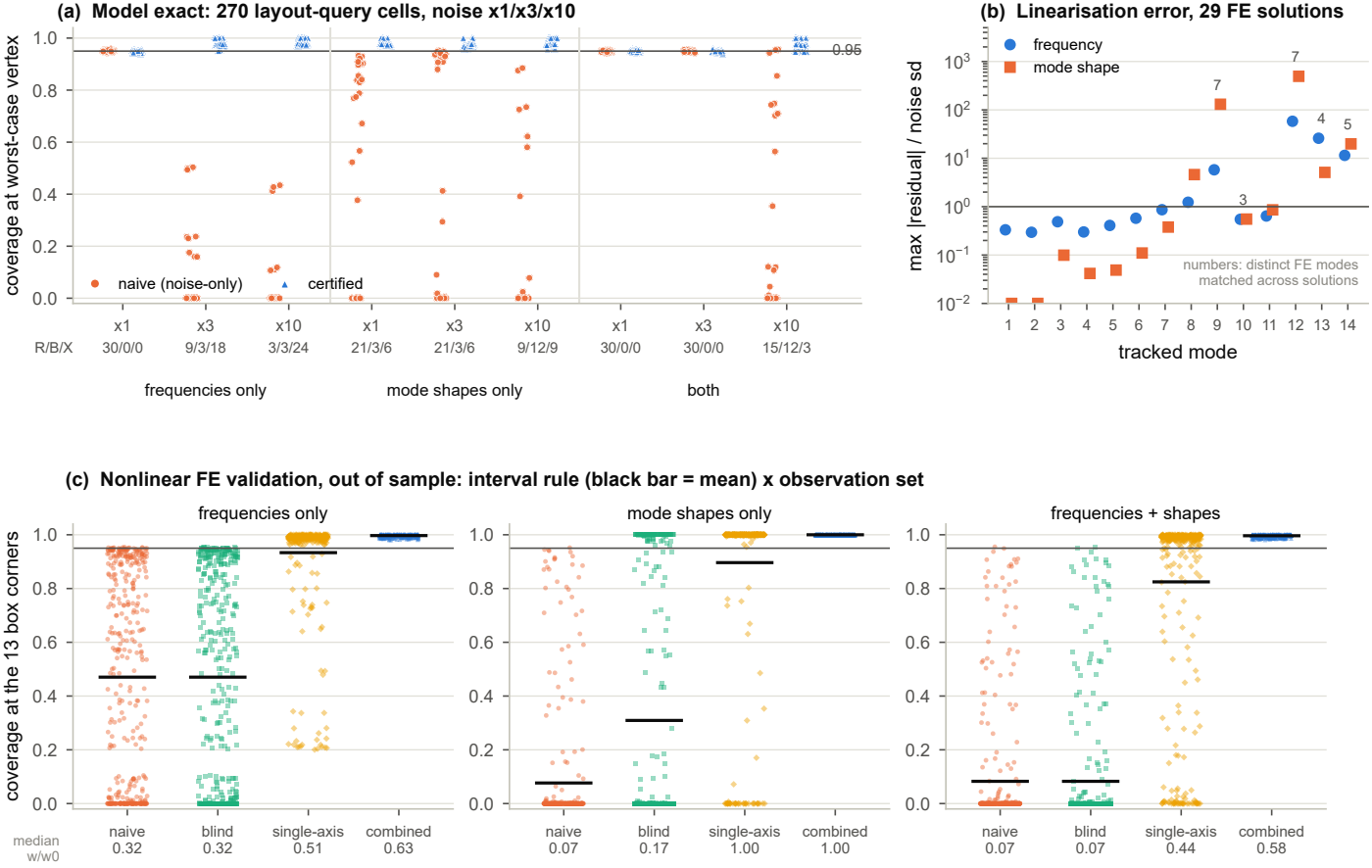

(a) Model exact: 270 layout-query cells, noise x1/x3/x10
coverage at worst-case vertex
1.0
0.8
0.6
0.4
0.2
0.0
0.95
naive (noise-only)
certified
x1
x3
x10
R/B/X
30/0/0
9/3/18
3/3/24
21/3/6
21/3/6
9/12/9
30/0/0
30/0/0
15/12/3
frequencies only
mode shapes only
both
(b) Linearisation error, 29 FE solutions
max |residual| / noise sd
frequency
mode shape
numbers: distinct FE modes
matched across solutions
tracked mode
(c) Nonlinear FE validation, out of sample: interval rule (black bar = mean) x observation set
frequencies only
mode shapes only
frequencies + shapes
coverage at the 13 box corners
naive
blind
single-axis
combined
median
w/w0
0.32
0.32
0.51
0.63
0.07
0.17
1.00
1.00
0.07
0.07
0.44
0.58